

Autonomous Fire Detection and Extinguishing Robot

Project Title: Fire Fighter Robot with Obstacle Avoidance

Course Name: Robotics Lab

Course Code: CSE 438

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Submitted To:

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Project Objective:

To develop an autonomous fire fighter robot that can detect flames, avoid obstacles, and automatically extinguish fire using a water pump system while reducing human involvement in hazardous environments.

Project Goals:

- Detect fire automatically using flame sensors
- Reduce human risk in dangerous fire situations
- Avoid obstacles during robot movement
- Automatically spray water on fire
- Demonstrate embedded system automation using Arduino
- Improve safety using autonomous robotic technology

Pseudocode:

```
START robot
INITIALISE flame sensors, ultrasonic sensor, motors, servo,
and water pump
WHILE robot is running
  READ flame sensor values
  READ obstacle distance using ultrasonic sensor
  IF obstacle detected nearby
    STOP robot
    TURN robot to avoid obstacle
  ELSE
    MOVE forward
    IF flame detected
      STOP robot
      MOVE toward fire source
      TURN ON water pump
      SPRAY water until fire disappears
      TURN OFF water pump
      CONTINUE searching for fire
    REPEAT process continuously
  END
```

Step-by-Step Algorithm:

Step 1: Start the robot program.

Step 2: Initialise flame sensors, ultrasonic sensor, motor driver, servo motor, and water pump.

Step 3: Continuously read flame sensor and obstacle distance values.

Step 4: If an obstacle is detected, stop the robot and change direction.

Step 5: If no obstacle is detected, move the robot forward.

Step 6: If flame is detected, stop the robot and move toward the fire source.

Step 7: Turn ON the water pump and spray water.

Step 8: Continue spraying until no flame is detected.

Step 9: Turn OFF the water pump.

Step 10: Repeat the whole process continuously.

Code:

```
#include <Servo.h> // (kept but not used to avoid pin con

// ----- Pin Map -----
const int ENA = 5;
const int IN1 = 7;
const int IN2 = 8;
const int IN3 = 9;
const int IN4 = 10;
const int ENB = 6;

const int TRIG = 4;
const int ECHO = 2;

const int FL_LEFT = A0;
const int FL_RIGHT = A1;

const int PUMP = 3;
const int LED_R = A2;
const int LED_G = A3;
const int LED_B = A4;

// ----- Tunables -----
int baseSpeed = 170;
int turnSpeed = 150;
int avoidDist = 20;
int flameThresh = 700;
int sprayMs = 4000;

// ----- Functions -----
void motorsStop() {
    analogWrite(ENA, 0);
    analogWrite(ENB, 0);
}
```

```
void motorsForward(int spdL, int spdR) {
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
    analogWrite(ENA, spdL);
    analogWrite(ENB, spdR);
}

void motorsBackward(int spdL, int spdR) {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, HIGH);
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, HIGH);
    analogWrite(ENA, spdL);
    analogWrite(ENB, spdR);
}

void motorsTurnLeft(int spd) {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, HIGH);
    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
    analogWrite(ENA, spd);
    analogWrite(ENB, spd);
}

void motorsTurnRight(int spd) {
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, HIGH);
    analogWrite(ENA, spd);
    analogWrite(ENB, spd);
}
```

Code:

```
long readUltrasonicCM() {
    digitalWrite(TRIG, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG, LOW);

    long dur = pulseIn(ECHO, HIGH, 30000);
    if (dur == 0) return 999;
    return dur / 58;
}

void pumpOn(bool on) {
    digitalWrite(PUMP, on ? LOW : HIGH);
}

void setLED(bool r, bool g, bool b) {
    digitalWrite(LED_R, r);
    digitalWrite(LED_G, g);
    digitalWrite(LED_B, b);
}

// 🔥 Stable flame detection (important fix)
bool detectFlame() {
    int sumL = 0, sumR = 0;

    for (int i = 0; i < 5; i++) {
        sumL += analogRead(FL_LEFT);
        sumR += analogRead(FL_RIGHT);
        delay(5);
    }

    int avgL = sumL / 5;
    int avgR = sumR / 5;

    return (avgL < flameThresh || avgR < flameThresh);
}
```

```
void setup() {
    Serial.begin(9600);

    pinMode(ENA, OUTPUT);
    pinMode(ENB, OUTPUT);
    pinMode(IN1, OUTPUT);
    pinMode(IN2, OUTPUT);
    pinMode(IN3, OUTPUT);
    pinMode(IN4, OUTPUT);

    pinMode(TRIG, OUTPUT);
    pinMode(ECHO, INPUT);

    pinMode(PUMP, OUTPUT);

    pinMode(LED_R, OUTPUT);
    pinMode(LED_G, OUTPUT);
    pinMode(LED_B, OUTPUT);

    pumpOn(false);
}

// ----- LOOP -----
void loop() {

    long dist = readUltrasonicCM();
    bool flame = detectFlame();

    Serial.print("Distance: ");
    Serial.print(dist);
    Serial.print(" Flame: ");
    Serial.println(flame);

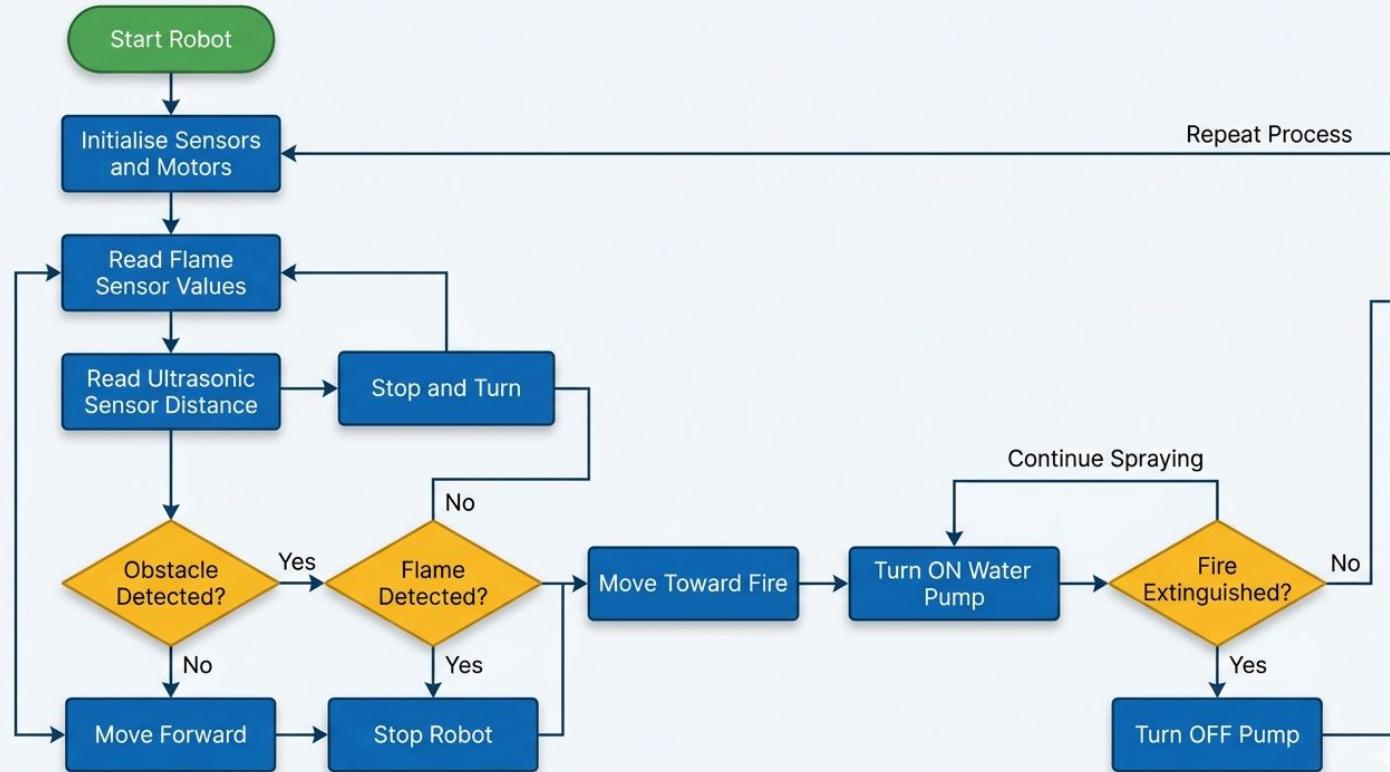
    // 🔥 FIRE MODE
    if (flame) {
        setLED(true, false, false); // RED

        motorsStop();
    }
}
```

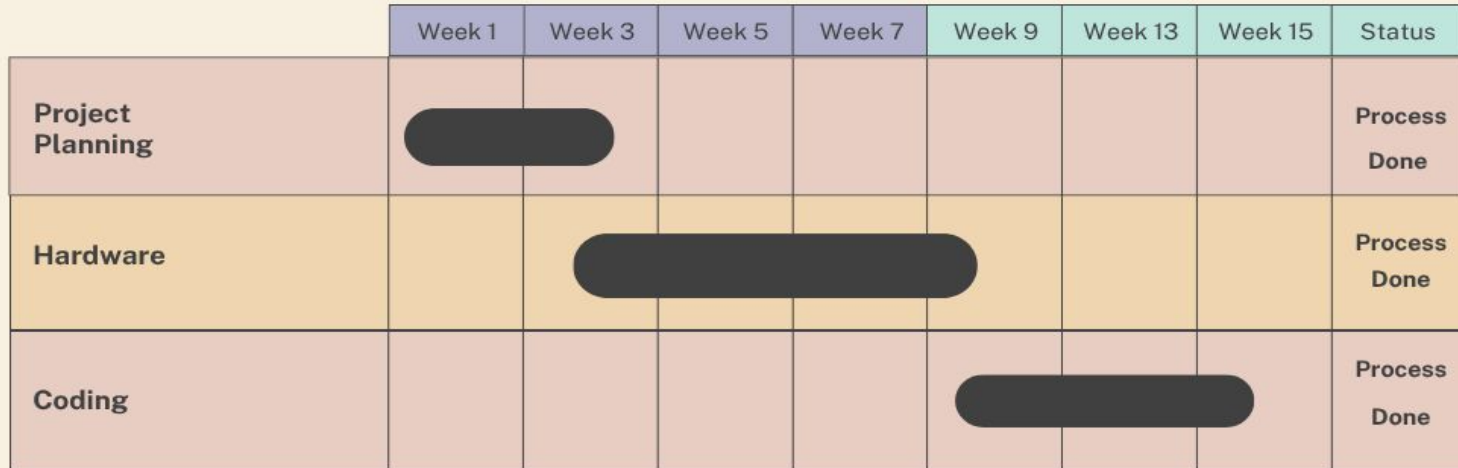
Code:

```
    motorsStop();  
    delay(300);  
}  
  
// 🚧 OBSTACLE MODE  
else if (dist < avoidDist) {  
    setLED(false, true, false); // GREEN  
  
    motorsBackward(150, 150);  
    delay(400);  
  
    motorsTurnRight(turnSpeed);  
    delay(500);  
}  
  
// 🚗 NORMAL MODE  
else {  
    setLED(false, false, true); // BLUE  
    motorsForward(baseSpeed, baseSpeed);  
}  
  
delay(50);  
}
```

Flowchart :



- # GRANTT CHART:



Executive Summary:

This project develops an autonomous fire fighter robot capable of detecting fire, avoiding obstacles, and extinguishing flames automatically using a water pump system. The robot uses flame sensors to identify fire and ultrasonic sensors to avoid collisions during movement. The main goal of the project is to improve safety, reduce human risk in dangerous environments, and demonstrate real-world embedded system automation using Arduino.

My Contribution in This Project:

My contribution was mainly in the hardware setup and coding section. I worked on:

Designing and assembling the robot hardware

Connecting flame sensors, ultrasonic sensor, motor driver, and water pump

Wiring and power supply management

Uploading and testing Arduino code

THANK YOU

Questions & Answers